

Modern Observable Inventory for Warped Extra-Dimensional Flavor Models

Scope and triage

A modern Randall–Sundrum flavor analysis should be organized by **operator classes**, not by a short list of “famous decays.” In today’s data, the observables that most directly probe warped flavor models are those sensitive to tree-level or one-loop effects from heavy KK vectors and fermions: **$\Delta F=2$ four-fermion operators, modified Z and W couplings, dipole operators, semileptonic FCNCs, and CP-odd dipoles such as EDMs**. That is why a credible 2026-era inventory must simultaneously cover kaon, $B_{d,s}$, charm, top, electroweak, charged-lepton-flavor-violation, and EDM observables. Recent measurements underscore this point: NA62 now quotes $\mathcal{B}(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = 9.6_{-1.8}^{+1.9} \times 10^{-11}$; HFLAV reports that R_K and R_{K^*} are now consistent with the SM while tensions in $b \rightarrow s \ell^+ \ell^-$ angular observables and rates remain; and MEG II has pushed $\mu \rightarrow e \gamma$ below 1.5×10^{-13} . ¹

For practical use in a warped-flavor paper, the RS model classes in your prompt can be mapped to observables as follows. This is a **phenomenology triage**, not a substitute for a full operator-level scan:

RS class	Observables that usually dominate
Anarchic RS	ε_K , K -mixing operators, $b \rightarrow s \gamma$, $B_{d,s}$ mixing and phases, $Z \rightarrow b \bar{b}$, ε'/ε , rare kaons, EDMs, and CLFV if leptons are also anarchic in the bulk
Custodial RS	Same flavor set as anarchic RS, but $Z b_L \bar{b}_L$ is partially protected, so rare semileptonic modes and B -mixing become comparatively more important than in non-custodial setups
RS-GIM	Same sectors, but many bounds soften; rare kaons, $b \rightarrow s \ell \ell$, dipoles, and precision electroweak flavor non-universality remain valuable
5D MFV RS	Down-sector FCNCs are often reduced; B -mixing, rare B decays, and electroweak observables remain relevant, but the classic anarchic ε_K crisis is weakened
Down-aligned / shining / flavor-aligned RS	K and $B_{d,s}$ FCNCs can be greatly reduced, so D^0 -mixing, rare charm, and top FCNCs become especially important
Up-aligned RS	Charm/top bounds soften, so the kaon and B-sector again become the dominant constraints
Lepton-bulk RS	$\mu \rightarrow e \gamma$, $\mu \rightarrow 3e$, $\mu N \rightarrow e N$, τ LFV, LFU tests, and semileptonic LFV meson decays become mandatory

RS class	Observables that usually dominate
CP-protected / CP-aligned RS	EDMs and CP-violating observables such as ε_K , $S_{\phi K_S}$, and B_s penguin phases are the place to test whether the protection actually works

The most useful working labels for a fit are:

- **Critical:** should be in the baseline likelihood now.
- **Important:** include unless you are deliberately narrowing the model space.
- **Secondary:** informative and often diagnostic, but presently weaker or theory-limited.
- **Future-facing:** not yet decisive, but likely to matter in the near term.

Kaon observables

The kaon sector is still the single most punishing arena for generic warped flavor. The reason is structural: RS models naturally generate new $\Delta S = 2$ operators, especially chirally enhanced LR structures, and modified Z -penguins that feed directly into rare kaons. Even in aligned or MFV-inspired versions, kaon observables remain the cleanest way to test whether down-sector FCNC protection is truly working.

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
Δm_K	Mass splitting in $K^0-\bar{K}^0$ mixing, $s \rightarrow d$, $\Delta S = 2$	Experiment is extremely precise; theory is less clean than for ε_K because long-distance contributions are important, so this is best treated as a conservative consistency bound rather than a sharpened short-distance constraint. FLAG supplies the key short-distance hadronic input $\hat{B}_K$. ²	Sensitive to the same KK-gluon / KK- Z four-quark operators that make anarchic RS difficult; especially relevant for anarchic, custodial, and generic RS-GIM constructions	Critical , but treat theory cautiously

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
ε_K	Indirect CP violation in $K^0-\bar{K}^0$ mixing, $s \rightarrow d$, $\Delta S = 2$	PDG quotes $\text{Re}(\varepsilon'/\varepsilon)$ separately and the kaon CP review gives the standard ε_K framework; modern lattice-based analyses still find that direct SM evaluations can show a sizable tension depending mainly on the choice of V_{cb} , while global CKM fits remain compatible with the SM. ³	Still the classic killer observable for anarchic RS because it is sensitive to CP-violating LR $\Delta S = 2$ operators; any claimed flavor protection in RS must be checked here first	Critical
ε'/ε in $K \rightarrow \pi\pi$	Direct CP violation in $s \rightarrow dq\bar{q}$ amplitudes	Experimental world average remains $\text{Re}(\varepsilon'/\varepsilon) = (16.6 \pm 2.3) \times 10^{-4}$. Theory has improved but is not yet settled; lattice and semi-analytic approaches continue to refine the SM prediction, and the uncertainty is still driven by hadronic matrix elements and isospin-breaking effects. ⁴	Especially important in RS because electroweak penguins and chromomagnetic operators can be enhanced; it is one of the best tests of warped down-sector CP structure beyond ε_K	Critical
$K^+ \rightarrow \pi^+ \nu \bar{\nu}$	Rare charged kaon decay, $s \rightarrow d\nu\bar{\nu}$	NA62 now reports $\mathcal{B} = (9.6^{+1.9}_{-1.8}) \times 10^{-11}$, in very good agreement with the PDG/SM expectation of about 8×10^{-11} . This is one of the cleanest short-distance flavor observables in the field. ⁵	RS models with modified Z or Z' couplings to down quarks can shift this decay at tree level; it is central for anarchic, custodial, RS-GIM, and also alignment models that do not perfectly protect $s \rightarrow d$ semileptonic currents	Critical

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
$K_L \rightarrow \pi^0 \nu \bar{\nu}$	CP-violating neutral kaon decay, $s \rightarrow d \nu \bar{\nu}$	Still not observed; PDG cites a current limit at the 10^{-9} level from KOTO, vastly above the SM expectation of order 3×10^{-11} . The SM prediction is extremely clean. ⁶	In RS this mode isolates CP-violating $s \rightarrow d Z^{(*)}$ couplings especially well; it is a decisive future classifier between CP-aligned and non-aligned scenarios	Critical , with a strongly future-facing experimental profile
$K_L \rightarrow \mu^+ \mu^-$	Rare neutral kaon decay, $s \rightarrow d \ell^+ \ell^-$	Measured precisely, with the absorptive two-photon contribution nearly saturating the rate; the short-distance dispersive piece is much harder and has seen recent theoretical refinement. ⁷	Valuable because warped models often modify the same semileptonic operators that feed $K \rightarrow \pi \nu \bar{\nu}$, but here the theory is less clean; still useful as a cross-check on Z -penguin patterns	Important
$K_S \rightarrow \mu^+ \mu^-$	Rare CP-sensitive kaon decay, $s \rightarrow d \mu^+ \mu^-$	PDG quotes an upper limit in the 10^{-10} range; the SM branching ratio is much smaller and the channel is increasingly discussed as an interference-based probe of short-distance physics. ⁸	Particularly interesting for RS models with right-handed semileptonic currents; presently not competitive with $K \rightarrow \pi \nu \bar{\nu}$, but it probes a complementary operator combination	Secondary , becoming future-facing
$K_L \rightarrow \pi^0 e^+ e^-$, $K_L \rightarrow \pi^0 \mu^+ \mu^-$	Rare neutral kaon semileptonic decays, $s \rightarrow d \ell^+ \ell^-$	Still limit-dominated experimentally; PDG quotes SM expectations at the $\sim 10^{-11}$ level, with known separation into direct-CPV, indirect-CPV, and CP-conserving pieces. ⁸	These modes are valuable if one wants to distinguish left-versus right-handed semileptonic operators in warped scenarios; they are not as clean or strong as $K \rightarrow \pi \nu \bar{\nu}$, but they help with operator diagnosis	Important

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
LFV kaon decays such as $K_L \rightarrow \mu e$, $K^+ \rightarrow \pi^+ \mu e$	Simultaneous quark and lepton flavor violation	Null tests with very strong upper limits compiled by PDG; the SM background is effectively zero. ⁶	Mandatory if the model includes nontrivial bulk lepton flavor or quark-lepton flavor linkages; usually subleading to $\mu \rightarrow e \gamma$ and $\mu N \rightarrow e N$, but very useful cross-checks	Secondary for quark-only RS, Important for lepton-bulk RS

For the kaon sector, the **minimum modern RS baseline** is: ε_K , ε'/ε , $K^+ \rightarrow \pi^+ \nu \bar{\nu}$, $K_L \rightarrow \pi^0 \nu \bar{\nu}$, and one semileptonic cross-check such as $K_L \rightarrow \mu^+ \mu^-$. If the construction claims down-sector alignment, then the kaon modes become the place where residual misalignment must be quantified rather than assumed away.

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Bottom observables

The B sector is the broadest modern handle on warped flavor because it probes **all** of the important RS operator classes: $\Delta B = 2$ mixing, electromagnetic and chromomagnetic dipoles, semileptonic Z -penguins, Higgs/scalar effects, right-handed currents, and modified charged currents. It is also where custodial protection and alignment can reshuffle which measurements dominate.

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
Δm_d , Δm_s	B_d and B_s mass splittings, $b \rightarrow d, s$, $\Delta B = 2$	HFLAV gives $\Delta m_d = 0.5069 \pm 0.0019 \text{ ps}^{-1}$ and $\Delta m_s = 17.766 \pm 0.006 \text{ ps}^{-1}$. SM interpretation depends mainly on lattice inputs $f_{B_q}^2 B_{B_q}$ and ξ . ¹⁰	Benchmark probes of KK-gluon and electroweak $\Delta B = 2$ operators; essential in anarchic, custodial, RS-GIM, and still necessary in MFV-like warped setups	Critical

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
$S_{J/\psi K_S}, C_{J/\psi K_S}, \phi_s$ from $B_s \rightarrow J/\psi \phi$	Mixing-induced CP phases in $b \rightarrow c\bar{c}s$ modes	HFLAV quotes $\phi_s^{c\bar{c}s} = -0.052 \pm 0.013$ rad; the “golden mode” $S_{J/\psi K_S}$ remains very close to the CKM expectation. ¹¹	These are the cleanest B -sector CP phases and are central for testing new weak phases in warped $\Delta B = 2$ operators; they are especially useful in CP-aligned vs non-aligned RS comparisons	Critical
$\Delta\Gamma_s$, semileptonic asymmetries $a_{sl}^{d,s}, q/p _{d,s}$	Width differences and CPV in mixing	HFLAV finds $\Delta\Gamma_s = 0.0781 \pm 0.0035 \text{ ps}^{-1}$, $ q_s/p_s = 1.0003 \pm 0.0014$, $ q_d/p_d = 1.0010 \pm 0.0008$, with no established CPV in mixing. ¹²	Not usually the first constraint, but they are important consistency checks when an RS fit introduces new $\Delta B = 2$ phases; useful for ruling out fine-tuned corners that survive $\Delta m_{d,s}$ alone	Important
$B \rightarrow X_s \gamma$	Inclusive radiative decay, $b \rightarrow s\gamma$	HFLAV quotes $\mathcal{B}(B \rightarrow X_s \gamma)_{E_\gamma > 1.6 \text{ GeV}} = (3.49 \pm 0.19) \times 10^{-4}$; current NNLO SM predictions are at $\sim 3.4 \times 10^{-4}$, so the agreement is tight. ¹³	A cornerstone bound on warped dipole operators from KK fermion loops. It is unavoidable in anarchic RS and remains important even when semileptonic FCNCs are partially aligned	Critical

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
Radiative CP observables $A_{CP}(B \rightarrow X_s \gamma)$, $S_{K^* \gamma}$, $B_s \rightarrow \phi \gamma$ CP asymmetries	CP and helicity structure in $b \rightarrow s \gamma$	Experimentally weaker than the inclusive branching ratio, but still an important null-test set; right-handed dipoles are far more visible here than in the total rate alone. HFLAV maintains dedicated radiative-decay averages. ¹⁴	Especially diagnostic for RS models with sizable right-handed currents or split LH/RH localization patterns	Important
Inclusive $B \rightarrow X_s \ell^+ \ell^-$	Inclusive semileptonic FCNC, $b \rightarrow s \ell^+ \ell^-$	Still theoretically cleaner than many exclusive channels; global status remains shaped by tensions in rates and angular observables rather than LFU ratios. HFLAV and LHCb reviews both emphasize that LFU anomalies have receded, but $b \rightarrow s \ell \ell$ tensions persist. ¹⁵	Important because RS often modifies C_9 , C_{10} , and their primed counterparts through Z -penguin physics; inclusive modes help distinguish genuine short-distance effects from exclusive hadronic issues	Critical
Exclusive $B \rightarrow K^{(*)} \ell^+ \ell^-$, including angular observables such as P'_5 , A_{FB} , F_L	Exclusive semileptonic FCNC, $b \rightarrow s \ell^+ \ell^-$	HFLAV reports persistent tensions in branching fractions and angular observables; LHCb's 2024–2026 amplitude analyses do not favor anomalously large long-distance charm effects, though a full SM interpretation is still under active study. ¹⁶	Among the best operator-diagnostic observables for warped Z -penguins and right-handed currents; they are especially valuable in custodial constructions where semileptonic currents can be prominent	Critical

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
R_K, R_{K^*}	LFU ratios in $b \rightarrow s \ell^+ \ell^-$	HFLAV now quotes $R_K = 0.947 \pm 0.047$ and $R_{K^*} = 1.028 \pm 0.074$ in the standard low- q^2 region, consistent with the SM. ¹²	No longer an anomaly-driven constraint, but still mandatory for lepton-bulk RS or any model with non-universal lepton localization; they cleanly test whether RS effects are lepton universal	Important for lepton-bulk RS, Secondary otherwise
$B_s \rightarrow \mu^+ \mu^-$, $B_d \rightarrow \mu^+ \mu^-$, and the $B_s \rightarrow \mu^+ \mu^-$ effective lifetime	Rare leptonic decays, $b \rightarrow s, d \ell^+ \ell^-$	Recent theory work quotes an SM $B_s \rightarrow \mu^+ \mu^-$ prediction around 3.64×10^{-9} ; current averages are close, with the measured value still somewhat lower but statistically compatible. ¹⁷	Very important in RS because these decays are directly sensitive to axial-vector and scalar semileptonic operators; they are among the cleanest $b \rightarrow s$ tests after B -mixing and $b \rightarrow s \gamma$	Critical
$B \rightarrow K \nu \bar{\nu}$, $B \rightarrow K^* \nu \bar{\nu}$	Invisible semileptonic FCNC, $b \rightarrow s \nu \bar{\nu}$	Belle II has reported new results, including evidence for $B^+ \rightarrow K^+ \nu \bar{\nu}$, and these decays remain theoretically clean, limited mostly by form factors. ¹⁸	In RS, $b \rightarrow s \nu \bar{\nu}$ shares the same semileptonic operator origin as $b \rightarrow s \ell^+ \ell^-$ but probes different lepton structures and invisible final states; very useful in custodial and lepton-bulk setups	Important , trending toward Critical

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
$B^+ \rightarrow \tau^+ \nu_\tau$	Purely leptonic charged-current decay, $b \rightarrow u \tau \nu$	Belle II reported evidence at 3.0σ with $\mathcal{B} = [1.24 \pm 0.41(\text{stat}) \pm 0.19(\text{syst})] \times 10^{-4}$. The SM prediction depends mainly on f_B and $ V_{ub} $. ¹⁹	Sensitive to modified charged-current couplings in RS, including non-universal W couplings and fermion mixing effects; also useful when connecting quark and lepton localization patterns	Important
$R(D), R(D^*)$	LFU ratios in $b \rightarrow c \tau \nu$	HFLAV gives $R(D) = 0.342 \pm 0.026$, $R(D^*) = 0.286 \pm 0.012$, with a combined tension of 3.14σ relative to the SM. ¹²	Not a “classic RS killer,” but important in models with modified left-handed charged currents or lepton non-universality; worth including if the model has nontrivial lepton localization or custodial fermion structure	Important
$B \rightarrow \pi K$ branching ratios and direct CP asymmetries	Nonleptonic EW-penguin and QCD-penguin system, mainly $b \rightarrow s q \bar{q}$	HFLAV continues to report the long-lived “ $K\pi$ CP puzzle,” with precise asymmetries such as $A_{CP}(B^0 \rightarrow K^+ \pi^-) = -0.0836 \pm 0.0032$. Theory is hadronic and less clean than in rare decays. ¹²	Important because RS can generate sizeable electroweak penguins and right-handed contributions. These observables are best used as diagnostics , not as stand-alone precision fit anchors	Important, but theory-limited

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
$S_{\phi K_S}, S_{\eta' K_S}, B_s \rightarrow \phi\phi$ CP observables	Penguin-dominated CP asymmetries, mainly $b \rightarrow s\bar{s}s$	LHCb has made increasingly precise $B_s \rightarrow \phi\phi$ measurements, with no established deviation from the SM; B_d penguin CP asymmetries remain standard components of HFLAV/ PDG CP summaries. ²⁰	These are classic RS diagnostics for new phases in electroweak or gluonic penguins. They matter most if a model survives the cleaner B -mixing and rare-semileptonic constraints	Important
LFV B decays such as $B^0 \rightarrow \mu\tau, B^0 \rightarrow e\tau, B \rightarrow K\mu e$	Simultaneous quark and lepton flavor violation	HFLAV quotes, for example, $B(B^0 \rightarrow \mu^\pm \tau^\mp) < 1.2 \times 10^{-5}$ and $B(B^0 \rightarrow e^\pm \tau^\mp) < 1.6 \times 10^{-5}$. ¹²	Usually subleading to $\mu \rightarrow e\gamma$ and $\mu N \rightarrow eN$, but they become useful cross-checks in unified quark-lepton warped models	Secondary

If you need a **minimal but still modern B -sector RS core**, it is: $\Delta m_{d,s}, S_{J/\psi K_S}, \phi_s, B \rightarrow X_s \gamma$, inclusive and/or exclusive $b \rightarrow s \ell^+ \ell^-$, $B_s \rightarrow \mu^+ \mu^-$, one invisible mode family $B \rightarrow K^{(*)} \nu \bar{\nu}$, and at least one penguin-dominated CP set. ²¹

Charm, top, and electroweak observables

This is the sector that becomes decisive as soon as you suppress down-sector FCNCs. In particular, **down-aligned, shining**, and some **5D MFV-like** constructions can survive kaon bounds only to run into D -mixing or top-FCNC trouble.

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
D^0 mixing: $x, y, q/p , \phi, x_{12}, y_{12}, \phi_{12}$	$c \rightarrow u, \Delta C = 2$ mixing and CPV	HFLAV reports $x = (0.41 \pm 0.04)\%$, $y = (0.65 \pm 0.02)\%$, $ q/p = 0.994 \pm 0.016$, $\phi = (-2.6 \pm 1.2)^\circ$; no-mixing is excluded by more than 11σ , but CPV remains consistent with zero. ¹²	This is the central observable family for down-aligned or shining RS models, where kaon and B constraints are weakened but up-sector FCNCs remain	Critical in down-aligned/shining RS; Important otherwise
Charm CPV: $\Delta a_{CP}^{dir}, a_{CP}^{ind}, A_\Gamma$	Direct and indirect CPV in $c \rightarrow uq\bar{q}$ decays	HFLAV quotes $\Delta a_{CP}^{dir} = (-0.159 \pm 0.029)\%$ and $a_{CP}^{ind} = (-0.010 \pm 0.012)\%$. ¹²	Especially useful because CPV observables can be cleaner NP probes than the absolute mixing amplitudes in charm; RS phases surviving kaon protection often show up here	Important
$D^0 \rightarrow \mu^+ \mu^-$	Rare charm FCNC, $c \rightarrow u\mu^+ \mu^-$	LHCb sets $\mathcal{B}(D^0 \rightarrow \mu^+ \mu^-) < 3.1 \times 10^{-9}$ at 90% CL, the world's best limit. ²²	A null test with tiny SM expectations; not as constraining as D -mixing today, but excellent as a cross-check on up-sector semileptonic FCNCs in aligned warped models	Important
Rare semileptonic charm modes such as $D \rightarrow P\ell^+ \ell^-, D^0 \rightarrow h^+ h^- \ell^+ \ell^-$, and angular CP tests	Rare charm FCNCs, $c \rightarrow u\ell^+ \ell^-$	LHCb has already performed angular analyses in $D^0 \rightarrow \pi^+ \pi^- \mu^+ \mu^-$ and $D^0 \rightarrow K^+ K^- \mu^+ \mu^-$, and recent rare-charm reviews emphasize the growing sensitivity of this program. ²³	These are not yet top-tier constraints, but they are very useful for operator diagnosis if a model predicts semileptonic up-sector FCNCs	Secondary , but distinctly future-facing

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
$t \rightarrow uZ, t \rightarrow cZ$	Top FCNC decays, $t \rightarrow qZ$	PDG's 2024 top review quotes 95% CL limits $\mathcal{B}(t \rightarrow Zu) < 6.2 \times 10^{-5}$ and $\mathcal{B}(t \rightarrow Zc) < 1.3 \times 10^{-4}$. ²⁴	One of the cleanest collider probes of warped up-sector FCNCs. If down-sector alignment is invoked, these modes often become decisive	Important now; Critical in aligned-up-sector studies
$t \rightarrow uH, t \rightarrow cH$	Top Higgs-FCNC decays, $t \rightarrow qH$	PDG quotes combined limits $\mathcal{B}(t \rightarrow uH) < 1.9 \times 10^{-4}$ and $\mathcal{B}(t \rightarrow cH) < 3.7 \times 10^{-4}$; the recent CMS analysis reaches essentially these scales. ²⁵	Highly relevant in warped models where Higgs FCNCs arise from fermion mixing and Yukawa misalignment; complementary to $t \rightarrow qZ$ because the operator origin is different	Important
$t \rightarrow q\gamma, t \rightarrow qg$	Top dipole FCNC decays	PDG quotes limits such as $\mathcal{B}(t \rightarrow u\gamma) < 0.85 \times 10^{-5}$, $\mathcal{B}(t \rightarrow c\gamma) < 4.2 \times 10^{-5}$, $\mathcal{B}(t \rightarrow ug) < 0.61 \times 10^{-4}$, and $\mathcal{B}(t \rightarrow cg) < 3.7 \times 10^{-4}$. ²⁴	Secondary today, but directly related to the same dipole structures that matter in $b \rightarrow s\gamma$; useful for global consistency checks	Secondary
$Z \rightarrow b\bar{b}: R_b, A_b, A_{FB}^{0,b}$	Flavor non-universal electroweak couplings of the b quark	PDG's electroweak review gives $R_b = 0.21629 \pm 0.00066$ versus an SM fit 0.21583 ± 0.00002 , and notes that the b -asymmetry extraction of $\sin^2 \theta_W$ remains somewhat high relative to the global fit. ²⁶	This remains a signature observable for warped models because fermion localization directly distorts $Zb\bar{b}$ couplings; custodial protection helps, but any realistic RS scan should still monitor this sector	Critical in non-custodial RS; Important in custodial RS

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
LFU and universality tests in τ decays and charged currents	W -coupling universality, lepton flavor non-universality	HFLAV gives $g_\tau/g_\mu = 1.0016 \pm 0.0014$, $g_\tau/g_e = 1.0018 \pm 0.0014$, and $g_\mu/g_e = 1.0002 \pm 0.0011$, all consistent with universality. ¹²	Usually subleading for quark-only RS, but important once lepton profiles are non-universal or the model perturbs charged currents appreciably	Secondary in quark-only RS, Important in lepton-bulk RS

For the alignment classes in your prompt, this section is the one that changes the game: **down alignment suppresses kaon and B FCNCs but elevates D -mixing and top FCNCs from “interesting” to “mandatory.”** ²⁷

Lepton flavor and CP observables

If the lepton sector is in the bulk, the fit cannot stop with quark flavor. In that case, the real “golden modes” are the three classic muon channels, supplemented by τ LFV, LFV Higgs and Z decays, and EDMs. This sector is also the best way to test whether a proposed CP-alignment mechanism actually suppresses all dangerous phases rather than only the hadronic ones.

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
$\mu^+ \rightarrow e^+ \gamma$	Charged LFV dipole decay	MEG II now reports a 90% CL upper limit $< 1.5 \times 10^{-13}$. The SM with neutrino masses is effectively zero on this scale. ²⁸	The canonical dipole constraint on lepton-bulk RS. If the lepton Yukawa sector is even mildly anarchic, this mode is often one of the most severe bounds	Critical
$\mu^+ \rightarrow e^+ e^- e^+$	Charged LFV four-fermion and dipole decay	The current best limit remains $< 1.0 \times 10^{-12}$ at 90% CL from SINDRUM; Mu3e is designed to improve by several orders of magnitude. ²⁹	Complementary to $\mu \rightarrow e \gamma$: it is especially valuable when four-fermion operators compete with dipoles, which is common in warped setups with heavy vector exchange	Critical

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
$\mu^- N \rightarrow e^- N$ conversion	Coherent muon-electron conversion in nuclei	Current best limit is at the $\mathcal{O}(10^{-13})$ level from SINDRUM II on gold; Mu2e aims for $\lesssim 8 \times 10^{-17}$ sensitivity. ³⁰	Often the most powerful probe of lepton-bulk RS because it receives coherent enhancement and is sensitive to dipole, vector, and scalar operator combinations	Critical
$\tau \rightarrow \mu\gamma, \tau \rightarrow e\gamma, \tau \rightarrow 3\mu, \tau \rightarrow 3e$	Tau LFV decays	HFLAV's 2025 tau summary emphasizes combined upper limits from Belle, BaBar, LHC experiments, and new Belle II inputs; all remain null so far. ³¹	These do not usually beat the muon channels, but they are indispensable for mapping flavor hierarchies in the lepton sector of warped models	Important
LFV Higgs decays $h \rightarrow \tau\mu, h \rightarrow \tau e$	Higgs flavor violation	ATLAS and CMS continue to find no signal; recent LHC searches keep the corresponding branching fractions below the per-mille to 10^{-3} – 10^{-2} level, depending on channel and combination. ³²	Particularly relevant in warped models because Higgs FCNCs can arise from the same profile misalignment that induces CLFV and top-Higgs FCNCs	Important
LFV Z decays $Z \rightarrow e\mu, Z \rightarrow e\tau, Z \rightarrow \mu\tau$	Gauge-boson LFV	Present collider limits are still weaker than the muon golden channels, but these modes remain conceptually important probes of non-universal bulk gauge couplings and leptonic mixing. ³³	Usually secondary in today's fits, but worth monitoring in custodial or strongly non-universal lepton constructions	Secondary

Observable	What is measured and transition	Current status and SM theory status	Why it matters for RS	Priority
Semileptonic LFV meson decays such as $K_L \rightarrow \mu e$, $K \rightarrow \pi \mu e$, $B \rightarrow K \mu e$, $B \rightarrow \mu \tau$	Joint quark-lepton flavor violation	Kaon modes have extremely strong limits from PDG compilations, while HFLAV tracks some B -sector LFV averages and limits. ³⁴	They are rarely the dominant bounds, but they provide a powerful consistency check once quark and lepton flavor symmetries are tied together in the same 5D construction	Secondary
Electron EDM d_e	CP-odd dipole in the electron sector	A 2023 result improved the previous best limit by about a factor of 2.4, pushing sensitivity into the $10^{-30} e \text{ cm}$ regime; earlier ACME results had already set the standard benchmark. ³⁵	Essential for any CP-protection claim in warped models. If CP alignment is not truly robust, d_e is often fatal	Critical
Neutron EDM d_n	CP-odd hadronic dipole observable	The best direct bound is still at the $10^{-26} e \text{ cm}$ level. ³⁶	A classic warped-flavor CP constraint, complementary to d_e because the operator mapping is different; central for testing whether a model solves the RS CP problem rather than just moving it	Critical

For any **bulk-lepton** or quark-lepton unified warped model, the baseline should include **all three muon golden channels plus d_e** . Tau LFV and LFV Higgs decays should then be added as flavor-structure diagnostics. ³⁷

Theory inputs and a recommended baseline dataset

A modern RS constraints paper does not only need observables; it also needs a clear **theory-input backbone** and a disciplined decision about which observables enter the **core likelihood** and which are handled as diagnostics or projections.

The authoritative input hierarchy should start with **HFLAV** for heavy-flavor averages and online updates, **PDG** for kaons, electroweak precision, top, and null-test compilations, and **FLAG** for lattice-QCD hadronic inputs such as $\hat{B}_K$, $f_{B_q} \sqrt{B_{B_q}}$, ξ , f_B , and form-factor averages whenever available. ³⁸

The main theory ingredients that a fit will need are these:

Observable families	Theory inputs that must be under control
K -mixing and ε_K	$\hat{B}_K$, CKM inputs, perturbative short-distance coefficients, treatment of long-distance pieces
ε'/ε	$K \rightarrow \pi\pi$ hadronic matrix elements, isospin breaking, electroweak penguins, chromomagnetic operators
$K \rightarrow \pi\nu\bar{\nu}$, $K_L \rightarrow \pi^0\ell\ell$, $K_{L,S} \rightarrow \mu\mu$	Short-distance semileptonic Wilson coefficients plus careful long-distance handling for the charged-lepton modes ⁷
$B_{d,s}$ mixing and leptonic decays	$f_{B_q}^2$, B_{B_q} , ξ , f_B , CKM tree inputs, $\Delta\Gamma_s$ conventions, and effective-lifetime treatment for $B_s \rightarrow \mu^+\mu^-$ ³⁹
$B \rightarrow X_s\gamma$, $B \rightarrow X_s\ell\ell$	NNLO perturbative inputs, consistent treatment of photon-energy cuts, HQE parameters
$B \rightarrow K^{(*)}\ell\ell$, $B_s \rightarrow \phi\mu\mu$, $B \rightarrow K^{(*)}\nu\bar{\nu}$	Form factors plus a documented treatment of nonlocal charm; LHCb's latest analyses are especially relevant here because they directly test whether enlarged hadronic effects can explain the data ⁴⁰
D -mixing and rare charm	For NP fits, matrix elements for $\Delta C = 2$ operators matter, but the dominant limitation is that the SM long-distance amplitude is not sharply predicted; CPV observables are therefore especially valuable
CLFV and $\mu N \rightarrow eN$	Dipole and four-fermion operator basis choice, nuclear overlap integrals, and consistent normalization conventions across targets
EDMs	Operator-basis matching from the high scale, hadronic and nuclear matrix elements for d_n , and a transparent treatment of electron-EDM versus semileptonic contributions

The **recommended baseline dataset** for a first modern RS flavor paper is therefore:

Core quark-sector baseline

ε_K , ε'/ε , $K^+ \rightarrow \pi^+\nu\bar{\nu}$, $K_L \rightarrow \pi^0\nu\bar{\nu}$, $\Delta m_{d,s}$, $S_{J/\psi K_S}$, ϕ_s , $B \rightarrow X_s\gamma$, one inclusive and one exclusive $b \rightarrow s\ell\ell$ family, $B_s \rightarrow \mu^+\mu^-$, $B \rightarrow K^{(*)}\nu\bar{\nu}$, $B \rightarrow \pi K$, one penguin-phase set such as $S_{\phi K_S}$ or $B_s \rightarrow \phi\phi$, D -mixing, $t \rightarrow qZ$, and $Z \rightarrow b\bar{b}$. ⁴¹

Add if down-sector alignment is claimed

Promote D -mixing, charm CPV, $D^0 \rightarrow \mu^+\mu^-$, and top FCNCs from “important” to “mandatory,” because these are the observables most likely to remain unsuppressed. ⁴²

Add if a bulk lepton sector is present

$\mu \rightarrow e\gamma$, $\mu \rightarrow 3e$, $\mu N \rightarrow eN$, τ LFV, LFU ratios, LFV Higgs decays, and d_e ; semileptonic LFV meson decays should then be used as cross-checks. ³⁷

Treat carefully rather than omit

Δm_K , $K_L \rightarrow \mu^+ \mu^-$, $B \rightarrow \pi K$, penguin-dominated hadronic CP asymmetries, and rare charm semileptonic modes. These are all useful, but they should not be assigned the same theory weight as $K \rightarrow \pi \nu \bar{\nu}$, ε_K , $B \rightarrow X_s \gamma$, or $B_s \rightarrow \mu^+ \mu^-$. ⁴³

Open questions and limitations

A few parts of the modern landscape remain intrinsically less settled than others. The most important limitations are the theory status of ε'/ε , long-distance contributions in $K_L \rightarrow \mu^+ \mu^-$, nonlocal charm effects in exclusive $b \rightarrow s \ell^+ \ell^-$, and the long-distance SM background in charm mixing and rare charm decays. Those observables should still be in the map, but not all of them should carry equal likelihood weight. ⁴⁴

The highest-confidence conclusion is straightforward: **a serious 2026-era RS flavor inventory is necessarily multi-sector**. For generic quark-sector RS it begins with kaons, B -mixing, dipoles, rare semileptonic B decays, $Z \rightarrow b\bar{b}$, and EDMs. For down-aligned or shining constructions it must elevate charm and top FCNCs. For bulk-lepton versions it must add the muon golden channels and electron EDM immediately. That is the observable map most likely to survive into a full numerical fit without major omissions. ⁴⁵

¹ ⁵ ⁹ ⁴¹ ⁴⁵ NA62 Collaboration refines measurement of rare particle decay – Home | CERN

https://home.cern/na62-collaboration-refines-measurement-rare-particle-decay/?utm_source=chatgpt.com

² 2025 update on ε_K in the Standard Model ...

https://arxiv.org/pdf/2503.00351?utm_source=chatgpt.com

³ ⁴ <https://pdg.lbl.gov/2024/reviews/rpp2024-rev-cp-violation.pdf>

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¹⁰ ¹¹ ¹² ¹³ ¹⁴ ¹⁵ ¹⁶ ²¹ ²⁷ ³⁸ ³⁹ ⁴² Results | Heavy Flavor Averaging Group

https://hflav.web.cern.ch/results?utm_source=chatgpt.com

¹⁷ Current Status of the Standard Model Prediction for the $B_s \rightarrow \mu^+ \mu^-$ Branching Ratio

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¹⁸ [2601.11279] Measurements of electroweak penguin and ...

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¹⁹ Measurement of $B^+ \rightarrow \tau^+ \nu_\tau$ branching fraction with a hadronic tagging method at Belle II

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